

Part 1 of 2

Plasma Arc Gasification For Waste Management

Plasma arc gasification is a relatively new technology for disposal of wastes. A large number of organisations, the world over, are developing plasma gasification systems for treating trash of various kinds and generating useful byproducts, while minimising the environmental pollution using proprietary processes

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Plasma gasification technology is particularly suitable for disposal of hazardous and radioactive wastes. It is now being seen as an attractive option for disposal of municipal solid wastes as well. This technology has an edge over the other processes involving combustion as it can turn solid municipal waste (organic as well as inorganic) into a clean, green, renewable fuel in the form of a synthesis gas (SynGas) with almost no residues that require further disposal.

The major advantages of plasma arc technology are:

1. Compared to combustion/incineration technology, it creates much less atmospheric pollution.

2. In techno-economic terms, oxides of nitrogen and sulphur are not emitted during normal operations because the system works in the absence of oxygen.

3. Toxic materials become encapsulated and are therefore much safer to handle than the toxic ash left by combustion/gasifier processes.

Despite the mentioned advantages, the process is cost-intensive. As per 12th Finance Commission guidelines for grants, plasma arc technology may only be employed for hazardous or

Other Well-known Waste Management Techniques

Managing domestic, industrial and commercial waste has traditionally consisted of collection, followed by disposal. Depending upon the type of waste and the area, a level of processing may follow collection. This processing may be to reduce the hazard of the waste, recover material for recycling, produce energy from the waste or reduce it in volume for more efficient disposal.

The other important waste disposal methods are briefed below. Of these, anaerobic digestion/biomethanation is comparatively a new method.

Landfills. Disposing of waste in a landfill is the most traditional method commonly practised in most countries. Few people want a landfill in their local neighbourhood because of adverse environmental impact.

Incineration. It is the process of destroying waste material by burning it. It involves high temperatures of 870°C to 1200°C (1400°F to 2200°F), to volatilise and combust (in the presence of oxygen). It is recognised as a practical method of disposing of hazardous waste materials (such as biological medical waste). The levels of chemical compounds released into the atmosphere when burning mixed waste, cause environmental pollution.

Anaerobic digestion/biomethanation. In this process, the organic fraction of the waste is segregated and fed into a closed container (biogas digester). In the digester, the segregated waste undergoes biodegradation in the presence of methanogenic bacteria under anaerobic conditions, producing methane-rich biogas and effluent.

The biogas can be used either for cooking/heating applications, or for generating motive power or electricity through dual-fuel or gas engines, low-pressure gas turbines or steam turbines. The sludge from anaerobic digestion, after stabilisation, can be used as a soil conditioner (manure) depending upon its composition, which is determined mainly by the composition of the input waste.

Biomethanation is the most widely used technology all over the world, more so in Europe. The main advantage is that it is environment-friendly and reliable. The only disadvantage is that it cannot treat non-biodegradable organic fractions.

bio-medical wastes. For various other well-known conventional waste management techniques, see Box.

What is plasma?

Plasma is the fourth state of matter (the other three states being solid, liquid and gas). It is an ionised gas that exists in nature, for example, dur-

ing lightning. Much of the universe exists in plasma state. The sun, stars and the interstellar matter are plasma manifestations. The gas in the upper parts of the ionosphere (say, above 400 km altitude) is completely ionised. On cosmological scales, plasma is entirely dominating the universe, with safe estimates giving that at least 99 per cent